

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 3
Total Marks:30

Duration : 45 Minutes

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I am OAVIKSHA P.L / 192572001 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

SIMATS ENGINEERING
SAVEETHA E OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO1 – ASSESSMENT TOOL 2
Constraint-Based Problem Solving

COURSE NAME: Artificial Intelligence

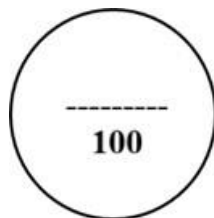
COURSE CODE: CSA17

QUESTIONS: 3

Duration: 1 Hour

Total Marks: 30

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Problem Understanding	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Constraint Analysis	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Solution Development	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Application of Concepts	15% (1.5)	/1.5	/1.5	/1.5	/4.5
Justification	10% (1)	/1	/1	/1	/3
Total per Question	10 Marks	/10	/10	/10	/30



Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Problem Analysis

The objective is to assign three doctors (D1, D2, D3) to three different shifts (Morning, Afternoon, Night) while satisfying all scheduling constraints. Since multiple conditions must be satisfied simultaneously, this problem is modeled as a Constraint Satisfaction Problem (CSP). Backtracking Search is selected because it systematically explores possible assignments and rejects any assignment that violates the constraints.

CSP Representation

Variables:

$\{D1, D2, D3\}$

Domain:

$\{Morning, Afternoon, Night\}$

Constraints

Constraint ID	Condition
C1	D1 cannot work Night shift
C2	D2 must be scheduled before D3
C3	Only one doctor can be assigned to one shift
C4	D3 cannot work Morning shift
C5	Every doctor must receive exactly one shift

Constraint Evaluation

Before assigning shifts, each constraint is analyzed.

- **C1** eliminates Night from D1's possible assignments.
- **C2** restricts the order of D2 and D3, reducing possible combinations.
- **C3** prevents duplicate assignments for any shift.
- **C4** removes Morning from D3's domain.
- **C5** guarantees a complete schedule where every doctor is assigned exactly one shift.

After applying these constraints, the search space becomes significantly smaller, allowing Backtracking Search to reach a solution efficiently.

Backtracking Search Process

Step 1

Assign

D1 → Morning

Constraint Check

- D1 is not assigned Night
- Shift is available

Current Assignment

Doctor	Shift
D1	Morning

Step 2

Assign

D2 → Afternoon

Constraint Check

- Afternoon is available
- D2 is before D3

Current Assignment

Doctor	Shift
D1	Morning
D2	Afternoon

Step 3

Remaining shift

Night

Assign

D3 → Night

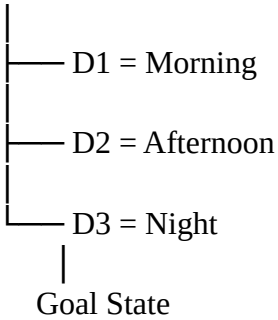
Constraint Check

- D3 is not Morning
- D2 works before D3
- Shift is unique
- Every doctor assigned once

Goal state reached.

Search Path

Start



Constraint Verification

Constraint	Verification	Status
D1 ≠ Night	D1 = Morning	✓
D2 before D3	Afternoon < Night	✓
One doctor per shift	All shifts are unique	✓
D3 ≠ Morning	D3 = Night	✓
Every doctor assigned once	Exactly one shift assigned	✓

Final Solution

Shift	Assigned Doctor
Morning	D1
Afternoon	D2
Night	D3

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Problem Analysis

A robot must navigate from the Start (S) position to the Goal (G) in a 5×5 grid. The environment contains obstacles that block movement. The objective is to determine the shortest path from the start to the goal using the Breadth-First Search (BFS) algorithm.

Grid Representation

+-----+				
S	.	.	X	.
+-----+				
.	X	.	X	.
+-----+				
.	.	.	.	.
+-----+				
X	X	.	X	.
+-----+				
.	.	.	.	G
+-----+				

Notation

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

Constraint Analysis

Constraint i: Movement Allowed (Up, Down, Left, Right)

The robot can move only in four directions. Diagonal movement is not permitted. This limits the search space and ensures BFS explores only valid neighboring cells.

Constraint ii: Cost of Each Move = 1

Every movement has the same cost. Since all edges have equal weight, BFS guarantees the shortest path in terms of the minimum number of moves.

Constraint iii: Cannot Pass Through Obstacles

Cells marked as obstacles are inaccessible. BFS ignores these nodes while expanding the search.

State Space Representation

- **Initial State:** Start (S)
- **Goal State:** Goal (G)
- **Operators:** Up, Down, Left, Right
- **Path Cost:** 1 per move

Breadth-First Search Execution

Step 1:

Start Node

[Queue = [S]

Visited = {S}]

Expand S

New nodes discovered:

[(0,1)

(1,0)]

Updated Queue:

[(0,1), (1,0)]

Step 2:

Expand **(0,1)**

Possible moves

- Right → (0,2)
- Down → Obstacle
- Left → Visited

Updated Queue

$[(1,0), (0,2)]$

Step 3:

Expand **(1,0)**

Possible moves

- Down $\rightarrow (2,0)$

Updated Queue

$[(0,2), (2,0)]$

Step 4:

Expand **(0,2)**

Possible moves

- Down $\rightarrow (1,2)$

Updated Queue

$[(2,0), (1,2)]$

Step 5:

Expand **(2,0)**

Possible moves

- Right $\rightarrow (2,1)$

Updated Queue

$[(1,2), (2,1)]$

Step 6:

Expand **(1,2)**

Possible moves

- Down $\rightarrow (2,2)$

Updated Queue

$[(2,1), (2,2)]$

Step 7:

Expand **(2,1)**

No new valid nodes.

Queue

[(2,2)]

Step 8:

Expand **(2,2)**

Possible moves

- Right → (2,3)
- Down → (3,2)

Queue

[(2,3), (3,2)]

Step 9:

Expand **(2,3)**

Possible moves

- Right → (2,4)

Queue

[(3,2), (2,4)]

Step 10:

Expand **(3,2)**

Possible moves

- Down → (4,2)

Queue

[(2,4), (4,2)]

Step 11:

Expand **(2,4)**

Possible moves

- Down → (3,4)

Queue

[(4,2), (3,4)]

Step 12:

Expand **(3,4)**

Possible moves

➤ Down → Goal

Queue

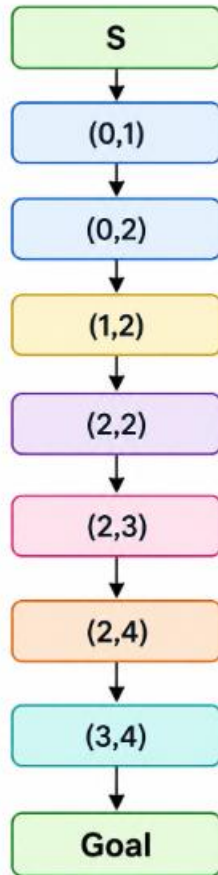
[Goal]

Goal Found.

Node Expansion Order:

Step	Expanded Node
1	S
2	(0,1)
3	(1,0)
4	(0,2)
5	(2,0)
6	(1,2)
7	(2,1)
8	(2,2)
9	(2,3)
10	(3,2)
11	(2,4)
12	(3,4)
13	Goal

Optimal Path



Path Cost Calculation

Move	Cost
S $\rightarrow$ (0,1)	1
(0,1) $\rightarrow$ (0,2)	1
(0,2) $\rightarrow$ (1,2)	1
(1,2) $\rightarrow$ (2,2)	1
(2,2) $\rightarrow$ (2,3)	1
(2,3) $\rightarrow$ (2,4)	1
(2,4) $\rightarrow$ (3,4)	1
(3,4) $\rightarrow$ Goal	1

Why BFS is the Appropriate AI Algorithm

Breadth-First Search (BFS) explores the search space **level by level**, expanding all nodes at the current depth before moving to the next level. Since each move has an equal cost of **1**, BFS guarantees the **shortest path** from the Start node to the Goal node. During the search, invalid moves through obstacles are discarded, and previously visited nodes are not explored again, making the search complete and optimal for an unweighted grid.

Final Result

Parameter	Result
Search Algorithm	Breadth-First Search (BFS)
Initial State	Start (S)
Goal State	Goal (G)
Optimal Path	$S \rightarrow (0,1) \rightarrow (0,2) \rightarrow (1,2) \rightarrow (2,2) \rightarrow (2,3) \rightarrow (2,4) \rightarrow (3,4) \rightarrow G$
Total Path Cost	8
Solution Status	Goal Reached Successfully

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

1. The robot can move up, down, left, right
2. Some cells are blocked (obstacles) and cannot be traversed
3. The robot has limited sensing capability (can only observe adjacent cells)
4. Each movement has a cost of 1, but entering risky zones has an additional cost of 2
5. The robot must minimize total cost while ensuring safe navigation
6. The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

4Problem Analysis

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot begins at the Start (S) position and must reach the Goal (G) while navigating through blocked paths and risky areas. Since the robot has only partial knowledge of the environment and can observe only adjacent cells, it must continuously update its knowledge and make intelligent decisions. The primary objective is to reach the survivor with the minimum total cost while ensuring safe navigation.

(a) Constraint Analysis

Each constraint directly influences the robot's planning and decision-making process.

Constraint 1: Movement Allowed (Up, Down, Left, Right)

The robot is restricted to four-directional movement. Diagonal movement is not permitted, reducing the possible actions available at each step.

Effect on Decision Making

- Limits the branching factor.
- Simplifies state expansion during search.
- Ensures movement follows valid navigation rules.

Constraint 2: Obstacles Cannot Be Traversed

Certain rooms are blocked due to debris or collapsed structures.

Effect on Decision Making

- The robot must detect blocked paths.
- Invalid paths are removed from consideration.
- Alternative routes must be explored dynamically.

Constraint 3: Limited Sensing Capability

The robot can observe only adjacent rooms.

Effect on Decision Making

- The robot initially has incomplete knowledge.
- Decisions are based only on local observations.
- New information is incorporated as exploration continues.

This makes the problem an Online Search Problem.

Constraint 4: Movement Cost

- Normal movement cost = **1**
- Risky zone cost = **1 + 2 = 3**

Effect on Decision Making

- The robot must evaluate both distance and safety.
- A slightly longer safe path may be preferred over a shorter risky path.
- Cost optimization becomes an important factor.

Constraint 5: Minimize Total Cost

The objective is not merely reaching the survivor but doing so with minimum overall cost.

Effect on Decision Making

- Every possible path is evaluated based on cumulative cost.
- Low-cost paths receive higher priority.

Constraint 6: Dynamic Knowledge Update

As the robot explores, newly discovered information changes the map.

Effect on Decision Making

- Previously selected paths may become invalid.
- The robot continuously replans its route.
- Navigation adapts to the changing environment.

(b) Solution Development Using Online Uniform Cost Search

Considering the partially observable environment and varying movement costs, an **Online Uniform Cost Search (UCS)** approach is the most suitable.

Step 1: Initialize the Search

- Robot starts at S.
- Initial knowledge contains only adjacent cells.
- Frontier contains the start node.

Step 2: Sense Adjacent Cells

The robot scans neighbouring rooms and classifies them as

- Free room
- Obstacle
- Risky zone

The internal map is updated accordingly.

Step 3: Calculate Path Cost

For each neighbouring node,

$$\text{Total Cost} = \text{Current Cost} + \text{Movement Cost}$$

Where

- Normal room = 1
- Risky room = 3

Step 4: Select the Minimum-Cost Node

Uniform Cost Search always expands the node with the lowest cumulative cost.

Priority Queue Example

Node	Total Cost
A	1
B	3
C	4

The robot expands **Node A** first because it has the minimum cost.

Step 5: Update Knowledge

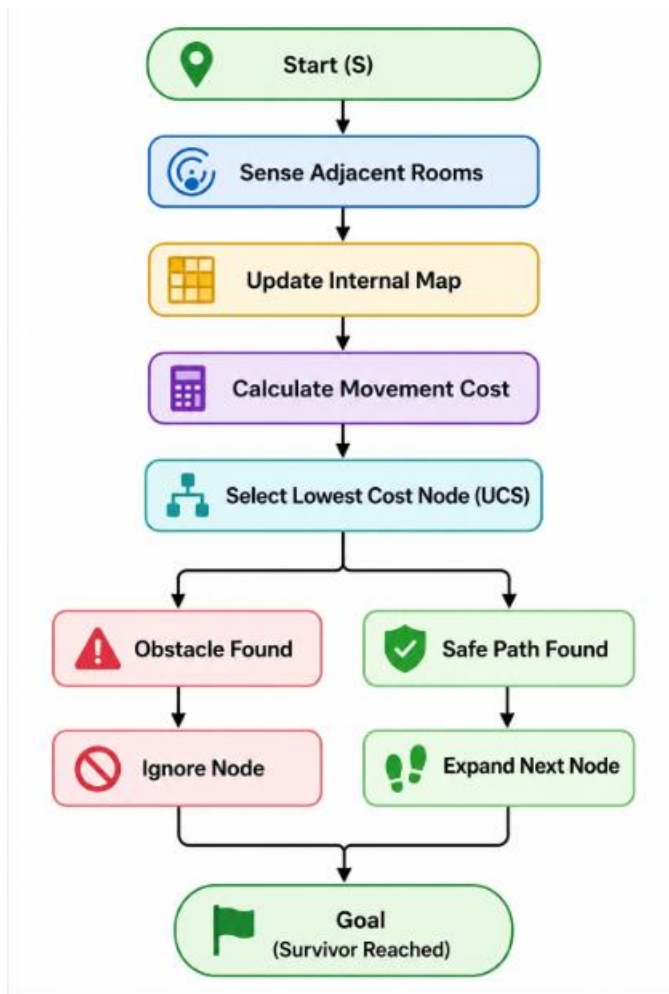
Whenever new rooms become visible,

- Update the internal map.
- Add newly discovered nodes to the priority queue.
- Ignore blocked cells.
- Recalculate the best route if necessary.

Step 6: Continue Until Goal

The search continues until the survivor (Goal) is reached with the minimum cumulative cost.

Workflow of the Proposed Solution



(c) Application of Artificial Intelligence Concepts

1. Intelligent Agent

The rescue robot functions as an **Intelligent Agent** because it:

- Perceives the environment through sensors.
- Processes available information.
- Selects appropriate actions.
- Learns new information during exploration.
- Achieves the rescue objective autonomously.

2. Rationality

A rational agent always chooses the action that maximizes its performance.

In this scenario, rational behaviour means:

- Avoid blocked rooms.
- Prefer safer paths.
- Reduce total movement cost.
- Reach the survivor efficiently.

Thus, every movement is selected based on maximizing the probability of successful rescue.

3. Environment Type

Environment Property	Type	Reason
Observability	Partially Observable	Robot senses only adjacent cells
Determinism	Deterministic	Actions produce predictable results
Dynamics	Dynamic	New information appears during exploration
State Space	Discrete	Building consists of separate rooms
Agents	Single Agent	Only one rescue robot performs the task

Performance Evaluation

Criterion	Outcome
Safety	High
Cost Optimization	High
Obstacle Handling	Effective
Dynamic Adaptation	Excellent
Goal Achievement	Successful

(d) Justification of the Selected Approach

The Online Uniform Cost Search algorithm is the most suitable because it effectively handles both partial observability and variable movement costs. Unlike Breadth-First Search, which considers only the number of moves, Uniform Cost Search evaluates the cumulative path cost, allowing the robot to avoid risky zones even if they offer a shorter route. The online search capability enables the robot to update its internal map as new information becomes available, making it well-suited for dynamic disaster environments. This approach ensures safe navigation, efficient resource utilization, and successful rescue operations while satisfying all the specified constraints.

Final Outcome

The rescue robot successfully applies Online Uniform Cost Search to navigate a partially observable disaster environment. By continuously sensing nearby rooms, updating its knowledge base, avoiding obstacles, minimizing movement costs, and selecting the least-cost path, the robot demonstrates the effective application of Artificial Intelligence principles, including intelligent agents, rational decision-making, and search-based problem solving. This approach ensures safe, adaptive, and optimal navigation to reach the survivor under the given constraints.